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Chapter 3

AUTONOMOUS UNDERWATER VEHICLE DYNAMICS

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ABSTRACT

The research on underwater systems has gained enormous attention during the last two decades because of applications taking place in many fields. Therefore, the significant number of Unmanned Underwater Vehicles (UUVs) has been developed for solving the wide range of scientific and applied tasks of ocean research and development in the world. Guidance, navigation, and control techniques are key research and development areas for the success of those sophisticated UUV missions.

One of the main objective of this chapter is to provides detailed explanations on the theory behind the main concepts that directly influence the design of the dynamic mathematical model of AUV and then to accomplish dynamic mathematical modeling of an AUV in MATLAB version 7.5 environment under different swimming conditions.

In order to develop high fidelity AUV simulation model and implement control and navigation algorithm for the vehicle, we need to know overall AUV modeling.

Another main focus of this chapter is to realize the parameter identification of hydrodynamic coefficients based on a Least Square Estimation (LSE) algorithm for a nonlinear mathematical modeling of AUV. Parameter Identification is very important to have the estimated values of these coefficients in order to accurately simulate the AUV's dynamic performance. The estimated hydrodynamic coefficients can be used as inputs not only for a mathematical model to analyze the maneuvering performance but also for the AUV's motion controller.

Keywords: Autonomous Underwater Vehicle, Dynamic Modeling, Parameter Identification, Least Square Estimation

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1. INTRODUCTION

Unmanned Underwater Vehicle (UUV) research began in the 1960s, with the first prototypes emerging in the 1980s. Nevertheless, the research on underwater systems has gained immense interest during the last two decades with applications taking place in multiple fields of marine systems.

Therefore, the significant numbers of UUVs have been developed for solving a wide range of scientific and applied tasks of ocean and seabed research and development in the world [1]. The military, as well as civilian industries can see great potential uses of UUVs in the underwater environment.

UUVs by definition are small submersible vehicles that contain independent propulsion systems and are capable of carrying sensors such as side-scan sonar, video cameras, depth sensor, and other oceanographic measuring devices [1]. UUVs are highly desirable as they can at least limit the level of human life risk and direct physical human involvement in a mission.

Typically, UUVs can be classified into two unmanned underwater systems that are Remotely Operated Vehicles (ROVs) and Autonomous Underwater Vehicles (AUVs): Each branch has its own pros and cons due to the mission type. ROVs, characterized by direct human assistance, is remotely operated through the presence of a tether cable; the other group is known as the AUVs, which is the topic of this chapter, characterized by their autonomous behavior, having cylindrical geometric shape and absence of a tether cable. This classification is of course not precise given the varying degrees of autonomy in both groups which can differ according to the requirements of the designed mission. In addition, the presence of a tether cable does not necessarily mean that the vehicle cannot perform autonomous tasks. The basic difference between AUVs and ROVs is that AUVs use “intelligence”, such as sensing and automatic decision-making. They have predefined plan of operations in its “mind” allowing them to perform tasks autonomously [1]. ROVs are remotely controlled by a human with the help of communication links based on tether cable (such as copper, fiber optic, etc.).

The AUVs have shown efficiency at performing complex search and inspection missions, and opening a number of new important application areas which include environmental monitoring, surveillance, underwater inspection of harbors and pipelines, geological and biological surveys, mine countermeasures, etc. [1].

Due to their versatility, compact size, independence and covertness, AUVs are a highly valuable asset in these environments. Possible missions for AUVs range from dedicated, rapid environmental assessment, special operations to reconnaissance, surveillance and intelligence. This wide range of possible applications and their partly contradictory nature calls for a large variety of vehicle configurations with regard to speed, range, depth, endurance, launch and recovery, payload and communication systems.

The nonlinear and multivariable dynamics, accompanied with parametric uncertainties and also immeasurable disturbances cause position control of an AUV to become an important issue. Dynamic equations of AUV are strongly nonlinear, multivariable and uncertain. Sea currents are disturbances for the system. Parameters of equations, which called hydraulic coefficients, should be estimated. Some of the parameters, such as hydrodynamic

coefficient, do not have accurate value. Computation and stimulation of these parameters is so complex and so time consuming and costly.

These parameters can vary due to environmental conditions. On the other hand, the control of AUV presents a number of unique and formidable challenges. An AUV will be required to move in adverse environments showing good performance and a great deal of autonomy under a variety of load conditions and unknown sea conditions. Moreover, AUV dynamic is highly nonlinear, coupled, time varying and subject to hydrodynamic uncertainties and external disturbances. For trajectory tracking, the problem statement requires the design of a depth and head control system for the vehicle in order to achieve precise trajectory [2].

In order to develop an accurate and robust navigation and control system, we need to know an overall mathematical modeling of AUV, which involves the interdisciplinary study of kinematics, hydrostatics, and hydrodynamics.

Hydrostatic is concerned with the equilibrium of underwater bodies at rest or moving with constant velocity, whereas hydrodynamics is concerned with bodies having accelerated motion [3]. In this chapter, we describe nonlinear dynamic model of an AUV for free swimming conditions.

Otherwise, in many literature, scientists employ parameter specific nonlinear AUV model for only one swimming condition.

Another main focus of this chapter is to realize the parameter identification of hydrodynamic coefficients based on Least Square Estimation (LSE) algorithm for a nonlinear mathematical modeling of AUV. The hydrodynamic coefficients of AUV can be estimated based on the calculations of the first principle of dynamics of AUVs and from statistical LSE of experimental data [4]. The identification of the parameters of the item being tracked can be approached in many different ways. The parameters can be determined once, from a model of the expected input signal. It could also be done adaptively utilizing a recursive method to identify the parameters of the incoming signal.

Another approach that estimates the hydrodynamic coefficients of AUVs is the observer method, in which a model-based estimation algorithm is used. A representative method amongst observer methods is the Kalman filter (KF), which has been widely used to estimate state variables and parameters [4]. Hwang [5] estimated the maneuvering coefficients of a ship and identified the dynamic system of a maneuvering ship using an extended Kalman filter (EKF) technique. Additionally, referencing [6], Meng and Veras, they concentrated on the application and comparison of EKF and iterated EKF for aerodynamic parameter estimation of a fixed wing UAV.

In another reference [7], Chowdhary compared the performance of three recursive parameter estimation algorithms for aerodynamic parameter estimation of two aircrafts derived from real flight data. These algorithms are the EKF, the simplified version of the Unscented Kalman filter (UKF) and the augmented version of the UKF.

On the other hand, the report [8] compares responses obtained by the KF, the least squares estimation, and the linear model for the NPS Phoenix AUV. The LSE provided results similar to those obtained by the KF, but the latter produced a more accurate model.

Referencing [9] Vandersteen, Rolain, Schoukens and Pintelon proposed a robust estimation algorithm for the estimation of static and nonlinear systems which can be described as a nonlinear function corrected with a rational form. The errors-in-variables-based algorithm solves the starting-value problem using an iterative, weighted least-squares procedure, which

constructs the rational form such that the set of normal equations becomes best conditioned, and uses a maximum-likelihood estimation step to increase the efficiency of the estimates.

The main focus of this chapter is to provide detailed explanations on the theory behind the main concepts that directly influence the design of the dynamic mathematical model of AUV and then to accomplish dynamic mathematical modeling of an AUV in MATLAB version 7.5 environment under different swimming conditions. Another main focus of this chapter is to realize the parameter identification of hydrodynamic coefficients based on the LSE algorithm for a nonlinear mathematical modeling of the AUV. It is important to have the estimated values of these coefficients in order to accurately simulate the AUV's dynamic performance. The estimated coefficients can be used as inputs not only for a mathematical model to analyze the maneuvering performance but also for an AUV's motion controller.

2. DYNAMIC MATHEMATICAL MODELING OF AN AUV

This section provides detailed explanations on the theory behind the main concepts that directly influence the design of the dynamic mathematical model of AUV.

In order to achieve robust and accurate navigation, guidance and control of an AUV, we require overall modeling of AUV, which is fairly complicated, and even when considered as a rigid body, an exact analysis is only possible by including the underlying infinite dimensional dynamics of the surrounding fluid [3]. This can be done using partial differential equations, which are solved by Computational Fluid Dynamics (CFD), but it still involves a formidable computational burden, which is infeasible for most practical applications.

As a result, the conventional approach has been to use finite-dimensional approximations.

On the other hand, AUV modeling involves the interdisciplinary study of kinematics, hydrostatics, and hydrodynamics. The study of hydrostatics is concerned with the equilibrium of underwater bodies at rest or moving with constant velocity, whereas hydrodynamics is concerned with bodies having accelerated motion [3].

An increased knowledge of hydrodynamic parameters then leads to a better navigational system design and performance on AUVs.

In this chapter we try to develop nonlinear mathematical modeling of an AUV for free swimming conditions and then as a case study, directly use the mathematical model of REMUS AUV that is designed to perform hydrographic analysis in the very shallow water.

REMUS is used for missions such as hydrographic surveys, mine counter-measure operations, harbor security operations, environmental monitoring debris field mapping, search and rescue operations, fishery operations, and scientific sampling and mapping.

2.1. Modeling Assumptions

2.1.1. Environmental Assumptions

The corresponding assumptions are made about the vehicle with respect to its environment [10]:

- The AUV is deeply submerged in a homogeneous and unbounded fluid. In other words, the AUV is located far from the free surface (no surface effects, i.e. no sea wave or vehicle wave-making loads), walls and bottom.
- The AUV does not experience underwater currents.

2.1.2. Dynamics Assumptions

In dynamic modeling of AUV, the following assumptions are used [11]:

- The AUV behaves as a rigid body of a constant mass.
- The earth's rotation is negligible for acceleration components of the vehicle's center of mass
- The primary forces that act on the AUV are inertial and gravitational in the center of buoyancy and are derived from hydrostatic, propulsion, thruster, and hydrodynamic lift and drag forces.
- The thruster assumption is that it uses an extremely simple propulsion model, which treats the vehicle propeller as a source of constant thrust and torque.

2.1.3. 6-DOF Rigid-Body Equations of Motion

AUVs move in six degrees of freedom (6-DOF) since six independent coordinates are necessary to determine the position and orientation angle of a rigid body dynamics. The first three coordinates and their time derivatives are based off of translational motion along the x , y and z -axes, while the last three coordinates (ϕ, θ, ψ) and their time derivatives are used to describe orientation angle and rotational motion [12].

Velocity and angular velocity components of the AUV relative to the body axes (x, y, z) are denoted by the velocity of surge, sway, heave motion, (u, v, w) and angular velocity of roll, pitch, and yaw motion (p, q, r) , respectively. X , Y , Z , K , M and N represent the resultant forces and moments with respect to the x , y , and z -axis. For AUVs, it is common to use the SNAME notation. In Table 1 below, the six different translational and rotational motion components are defined as: *surge*, *sway*, *heave*, *roll*, *pitch* and *yaw*, respectively [3].

2.1.4. Coordinate Frames

Typically, three different right-handed and rectangular coordinate frames are used for defining AUV motion, which is a 6 degrees of freedom (6-DOF). First, the body axes (x, y, z) have their origin at the center of buoyancy (CB) with x directed toward the bow along the hull centerline axis, y directed to the starboard side, and z toward the keel (Figure 2).

Table 1. AUV Dynamic Components [1]

Motion Components	Forces and Moments	Linear and Angular Velocities	Position and Euler Angles
Surge	X	u	X
Sway	Y	V	Y
Heave	Z	w	Z
Roll	K	P	ϕ

Pitch	M	Q	θ
Yaw	N	R	ψ

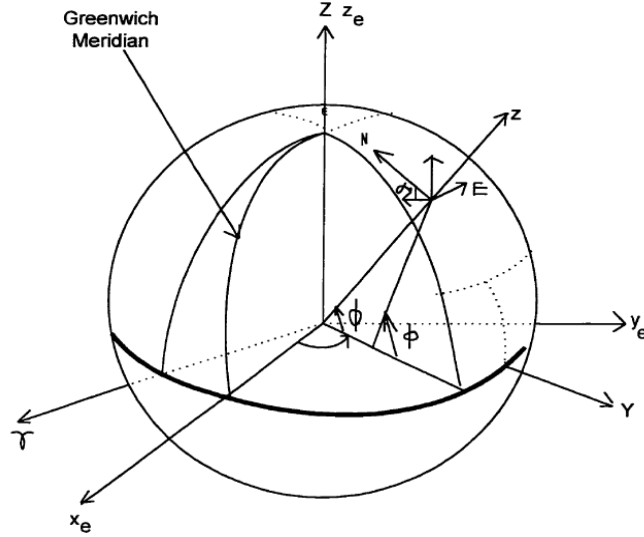


Figure 1. 6-DOF navigational frame [11].

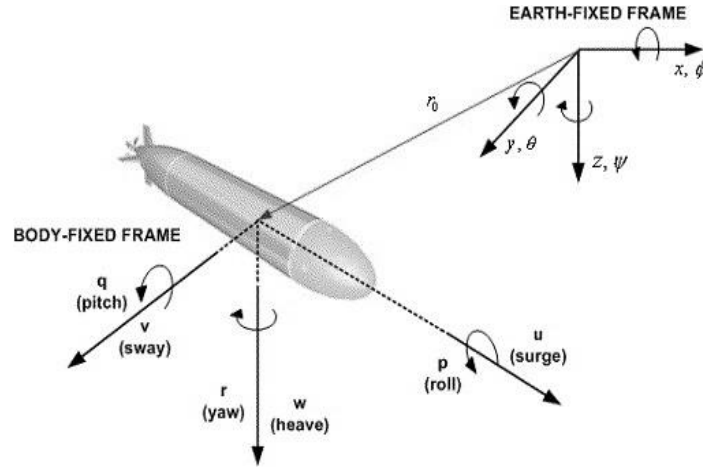


Figure 2. AUV body-fixed and earth-fixed coordinate system.

The axes fixed in the earth are (x_e, y_e, z_e) with the x_e, y_e plane in the water surface and z_e directed downward into the ocean. Second, the Earth-fixed coordinates frame is also measured to CB of AUV. If roll, pitch and yaw orientation angles ϕ, θ, ψ of the AUV are zero, the (x, y, z) axes will be parallel to the (x_e, y_e, z_e) axes, respectively.

The third coordinate frame (x_f, y_f, z_f) is fixed in the fluid, which can move with a constant velocity (u_f, v_f, w_f) relative to the earth-fixed frame. The x_f, y_f, z_f axes are always parallel to the x_e, y_e, z_e axes, respectively [3].

A set of axes commonly used with the Earth-fixed axis system is shown in 0, where X_e axis is chosen to point north, Y_e axis points east with the orthogonal triad being completed when Z_e axis pointing down.

A transformation matrix containing ‘Euler’ angles ϕ, θ, ψ where ϕ is roll, θ is pitch, and ψ is yaw, must be defined. More specifically, the adopted Euler angles move the reference frame to the referred body frame, following a Z-Y-X (or so-called 3-2-1) rotation sequence [13]. As a result, the transformation order from the earth-fixed frame ($\mathfrak{T}_e$) to the body-fixed frame ($\mathfrak{T}_b$) is given in the following equation:

$$\mathfrak{T}_e \xrightarrow[\psi]{u_3} \mathfrak{T}_1 \xrightarrow[\theta]{u_2} \mathfrak{T}_2 \xrightarrow[\phi]{u_1} \mathfrak{T}_b \quad (1)$$

Transformation matrix or direction cosine matrix is defined in the following equation:

$$T(\phi, \theta, \psi) = \begin{bmatrix} \cos \psi \cos \theta & \sin \psi \cos \theta & -\sin \theta \\ \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \cos \theta \sin \phi \\ \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (2)$$

Transformation from a global velocity vector to the local velocity vector is given by

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = T(\phi, \theta, \psi) \cdot \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} \quad (3)$$

The global angular velocity vector $[p, q, r]$ can be transformed into the rates of change of the Euler angles as given in Equation (4):

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (4)$$

Three dimensional (3D) spatial rotations can be parametrized using both Euler angles and unit quaternions. Unit quaternion provides a convenient mathematical notation for representing orientation angles and rotations of vehicle in three dimensions.

Compared to Euler angles, unit quaternions are simpler to compose and avoid the problem of the gimbal lock. Additionally, compared to rotation matrices, they are more numerically stable and may be more efficient [14].

Nevertheless, Euler angles are preferred in this chapter for simplification reasons.

After a general model structure for AUV is derived, we look further into the modeling aspects in terms of environmental disturbance models, in which waves (wind generated),

wind, and sea currents will be considered. In general, these environmental disturbances will both be additive and multiplicative to dynamic EOM [3].

Transformation order from body-fixed frame ($\mathfrak{T}_b$) to sea current frame ($\mathfrak{T}_w$) using orientation angle of α and β is given in following equation

$$\mathfrak{T}_b \xrightarrow[-\alpha]{u_2} \mathfrak{T}_1 \xrightarrow[\beta]{u_3} \mathfrak{T}_w \quad (5)$$

The transformation matrix from the body-fixed frame ($\mathfrak{T}_b$) to the sea current frame ($\mathfrak{T}_w$):

$$\begin{aligned} \hat{C}^{(b,w)} &= R_2(-\alpha) \times R_3(\beta) \\ \hat{C}^{(b,w)} &= \begin{bmatrix} \cos \alpha \cos \beta & -\cos \alpha \sin \beta & -\sin \alpha \\ \sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & -\sin \alpha \sin \beta & \cos \alpha \end{bmatrix} \end{aligned} \quad (6)$$

The sea current velocity components are assumed constant relative to axes fixed in the earth. This permits the use of axes fixed in the fluid as an inertial frame of reference. AUV velocity with respect to the sea is expressed as

$$\vec{V}_{b/w} = \vec{V}_{b/e} - \vec{V}_{w/e} \quad (7)$$

Then, dynamic pressure is calculated as

$$Q_d = \frac{1}{2} \rho V_{b/w}^2 \quad (8)$$

In this chapter, however, we ignore the sea currents, which normally effects the AUV motion. Additionally, wind and wind generated waves phenomena will not be discussed since the attention is focused on AUVs performing a motion or manipulation task in an underwater environment.

2.1.5. Newtonian and Lagrangian Mechanics

The EOM of AUVs are highly nonlinear, time-varying and coupled due to hydrodynamic added mass, lift, drag, coriolis and centripetal forces, which are acting on the vehicle and generally include uncertainties [3]. Overall 6-DOF nonlinear dynamic EOM can be expressed in the matrix form as

$$M(v)\dot{v} + C(v)v + D(v)v + g(\eta) = \tau \quad (9)$$

where, $M(v)$ is inertia matrix (including added mass), $C(v)$ is matrix of Coriolis and centripetal terms (including added mass), $D(v)$ is damping matrix, $g(\eta)$ is vector of gravitational forces and moments, τ is vector of control inputs, and $v = [u, v, w, p, q, r]^T$.

The coupled EOM of AUV are derived from two possible modeling approaches; one is a Lagrangian method and the other is a Newtonian-Euler formulation.

Basically, the Lagrangian approach consists of three main steps: first, to formulate a suitable expression for the vehicle's kinetic (T) and potential energy (P), second, to compute the Lagrangian L ($L=T-P$), and finally to apply the Lagrangian (L) to the Lagrangian formulation [12].

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) - \frac{\partial L}{\partial \eta} = J^{-T}(\eta) \tau \quad (10)$$

Another modeling approach is the Newtonian-Euler formulation, which is based on Newton's Second Law that relates mass (m), acceleration (a) and force ($\mathbf{F}$). Euler suggested expressing Newton's Second Law in terms of conservation of both linear and angular momentum. The forces ($\mathbf{F}$) and moments ($\mathbf{M}$) refers to the body's center of gravity [3].

In this study, the dynamic behavior of an AUV is described through Newton's laws of linear and angular momentum.

Newton's Second Law is expressed as:

$$\mathbf{F} = \frac{d}{dt} \{m\mathbf{V}\} \quad (11)$$

$$\mathbf{M} = \frac{d}{dt} \{\mathbf{H}\} \quad (12)$$

where $\mathbf{F}$ represents the sum of all externally applied forces, $\mathbf{M}$ represents the sum of all applied torques, and $\mathbf{H}$ is the angular momentum.

It is convenient to regard the sums of applied torque ($\mathbf{M}$) and force ($\mathbf{F}$) as consisting of an equilibrium point and a perturbational component. Thus, assuming constant AUV mass

$$\mathbf{F} = \mathbf{F}_0 + \Delta \mathbf{F} = m \frac{d}{dt} \{\mathbf{V}\} \quad (13)$$

$$\mathbf{M} = \mathbf{M}_0 + \Delta \mathbf{M} = \frac{d}{dt} \{\mathbf{H}\} \quad (14)$$

The subscript “₀” denotes the equilibrium condition, and “ Δ ” the component of perturbation. Equilibrium of AUV by definition must be an unaccelerated motion along a straight path; during this motion the linear velocity vector relative to fixed space is invariant, and the angular velocity is zero. Thus, both $\mathbf{F}_0$ and $\mathbf{M}_0$ are zero.

Furthermore, since the axis system being used as an inertial reference system is the Earth-fixed coordinate system, Equations (15) and (16) can be expressed as

$$\Delta \mathbf{F} = m \frac{d}{dt} \{\mathbf{V}_T\}_E \quad (15)$$

$$\Delta \mathbf{M} = \frac{d}{dt} \{\mathbf{H}\}_E \quad (16)$$

The force equation based on rate of change of velocity $\mathbf{V}$ relative to Earth's fixed axis frame is given by

$$\vec{F} = D_e(m\vec{V}) = D_b(m\vec{V}) + \vec{\omega}_{b/e} \times m\vec{V} \quad (17)$$

where $\omega_{b/e}$ is the angular velocity of the AUV with respect to the Earth fixed coordinate frame. The open form of Equation (17) is given by

$$\mathbf{F} \Rightarrow \begin{cases} F_x = X = m(\dot{u} + qw - rv) \\ F_y = Y = m(\dot{v} + ru - pw) \\ F_z = Z = m(\dot{w} + pv - qu) \end{cases} \quad (18)$$

After the rearrangement of $\mathbf{F}$, translational accelerations become:

$$\begin{aligned} \dot{u} &= \frac{F_x}{m} - qw + rv \\ \dot{v} &= \frac{F_y}{m} - ru + pw \\ \dot{w} &= \frac{F_z}{m} - pv + qu \end{aligned} \quad (19)$$

Similarly, after transforming from the body-fixed frame to the Earth-fixed frame the moment equation and its open form becomes

$$\begin{aligned} \vec{M} &= D_e(\hat{\mathbf{I}} \vec{\omega}_{b/e}) = D_b(\hat{\mathbf{I}} \vec{\omega}_{b/e}) + \vec{\omega}_{b/e} \times (\hat{\mathbf{I}} \vec{\omega}_{b/e}) \\ \vec{M} &\Rightarrow \begin{cases} M_x = K = I_x \dot{p} - I_{yz}(q^2 - r^2) - I_{zx}(\dot{r} + pq) - I_{xy}(\dot{q} - rp) - (I_y - I_z)qr \\ M_y = M = I_y \dot{q} - I_{zx}(r^2 - p^2) - I_{xy}(\dot{p} + qr) - I_{yz}(\dot{r} - pq) - (I_z - I_x)rp \\ M_z = N = I_z \dot{r} - I_{xy}(p^2 - q^2) - I_{yz}(\dot{q} + rp) - I_{zx}(\dot{p} - qr) - (I_x - I_y)pq \end{cases} \end{aligned} \quad (20)$$

For a rigid body, angular momentum can be defined as

$$\mathbf{H} = \mathbf{I}\boldsymbol{\omega} \quad (21)$$

where the inertia matrix is defined as

$$\mathbf{I} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \quad (22)$$

where I_{ii} denotes a moment of inertia, and I_{ij} a product of inertia $j \neq i$. In this chapter, we assume that the AUV is symmetrical along the XY and XZ planes, therefore cross inertia parameters become

$$I_{xy} = I_{yz} = I_{xz} = 0 \quad \text{and} \quad I_y = I_z \quad (23)$$

As a result of this symmetry, the rotational EOM of AUV in particular, becomes:

$$\begin{aligned} M_x &= I_x \dot{p} \\ M_y &= I_y \dot{q} - (I_z - I_x) r p \\ M_z &= I_z \dot{r} - (I_x - I_y) p q \end{aligned} \quad (24)$$

After rearrangement, Equation (24) is expressed as:

$$\begin{aligned} \dot{p} &= \frac{M_x}{I_x} \\ \dot{q} &= \frac{1}{I_y} [M_y + (I_z - I_x) r p] \\ \dot{r} &= \frac{1}{I_z} [M_z + (I_x - I_y) p q] \end{aligned} \quad (25)$$

Finally, the derivative of Euler angles is defined as:

$$\begin{aligned} \dot{\psi} &= \frac{q \sin \varphi + r \cos \varphi}{\cos \theta} \\ \dot{\theta} &= q \cos \varphi - r \sin \varphi \\ \dot{\varphi} &= p + (q \sin \varphi + r \cos \varphi) \tan \theta \end{aligned} \quad (26)$$

2.1.6. Gravitational Forces

Gravitational forces always exist in the AUV. It can be assumed that gravity acts at the center of gravity (CG) of the vehicle. Since the centers of mass and gravity coincide in an AUV, there is no external momentum produced by gravity on the CG.

Therefore, for the body axis system, gravity contributes only to the external force vector $\mathbf{F}$. Three components of the gravitational force in the body frame depend on the AUV's attitude relative to an inertia frame. The gravitational force acting upon an AUV is most obviously expressed in terms of the Earth's axes. With respect to these axes, the gravity vector mg , is directed along the Z_e axis. Figure 3 demonstrates the alignment of the gravity vector with respect to the body-fixed axes. In Figure 3, θ represents the pitch angle between the gravity vector and the $Y_b Z_b$ plane; the angle is positive when the nose of the AUV goes up, Φ represents the roll angle between Z_b axis and the projection of the gravity vector on the $Y_b Z_b$ plane. Direct solution of the vector mg into X , Y and Z components produces:

$$\begin{aligned} g_x &= mg \sin(-\theta) = -mg \sin \theta \\ g_y &= mg \cos(-\theta) \sin \Phi = mg \cos \theta \sin \Phi \\ g_z &= mg \cos(-\theta) \cos \Phi = mg \cos \theta \cos \Phi \end{aligned} \quad (27)$$

In general, the Euler angles (Φ , θ , Ψ) are not simply the integrals of the angular velocity p , q , and r (Equation (25)). It is necessary to relate these and their derivatives to the angular velocities p , q , and r . This depends upon whether the gravitational vertical seen from the AUV is fixed or whether it rotates relative to inertial space [16].

The manner in which the angular orientation and velocity of the body axis system with respect to the gravity vector is expressed by depending upon the angular velocity of the body axes about the vector mg . The external forces acting on the AUV can be expressed as:

$$\begin{aligned} X &= F_x + g_x \\ Y &= F_y + g_y \\ Z &= F_z + g_z \end{aligned} \quad (28)$$

where F_x , F_y , and F_z are the gravitational terms, and g_x , g_y , and g_z represent the hydrodynamic and thrust forces respectively.

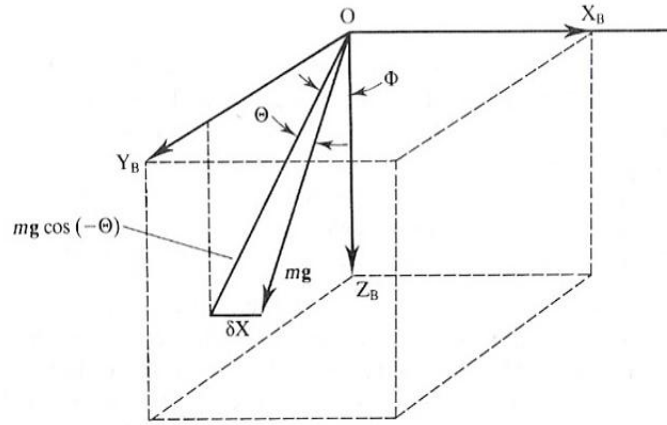


Figure 3. The orientation of gravity vector with respect to the body axis system [15].

Hence, from the Equations (13) and (15), the force equations become:

$$\begin{aligned}
 F_x &= ma_{x_{CG}} = m(\dot{u} + qw - rv + g \sin \theta) \\
 F_y &= ma_{y_{CG}} = m(\dot{v} + ru - pw - g \cos \theta \sin \Phi) \\
 F_z &= ma_{z_{CG}} = m(\dot{w} + pv - qu - g \cos \theta \cos \Phi)
 \end{aligned} \tag{29}$$

The gravitational acceleration forces and moments are represented by the weight minus buoyancy ($W-B$) and weight moment terms respectively [4]. In Figure 4, the Simulink model of gravitational force calculation is shown.

2.1.7. Hydrostatic Forces and Moments

When an AUV is submerged in a fluid under the effect of gravity, two forces act on the vehicle: the gravitational force, which is mentioned in the previous sub-section and the buoyancy, which is called “hydrostatic effect”.

The buoyant force acting on the center of buoyancy (CB) is represented in body-fixed frame (Figure 5). It can be recognized that the difference between gravity and buoyancy ($W-B$) only affects the linear force acting on the vehicle. It is also clear that the restoring linear force is constant in the Earth-fixed frame [11]. On the other hand, the two vectors of the first moment of inertia W and B affect the momentum acting on the vehicle and are constant in the body-fixed frame. A solid body submerged in a fluid will have upward buoyant force acting on it equivalent to the weight of displaced fluid, enabling it to float or at least appear to become lighter. If the buoyancy exceeds the weight, then the object floats; if the weight exceeds the buoyancy, the object sinks. If the buoyancy equals the weight, the body has neutral buoyancy and may remain at its level. Discovery of the principle of buoyancy, which is a result of the hydrostatic pressure in the fluid, is attributed to Archimedes [17].

After applying hydrostatic force coefficients to Equation 29, translational force equations become:

$$\left. \begin{aligned} F_x &= ma_{x_{CG}} = m(\dot{u} + qw - rv) + (W - B)\sin\theta \\ F_y &= ma_{y_{CG}} = m(\dot{v} + ru - pw) - (W - B)\cos\theta\sin\Phi \\ F_z &= ma_{z_{CG}} = m(\dot{w} + pv - qu) - (W - B)\cos\theta\cos\Phi \end{aligned} \right\} \quad (30)$$

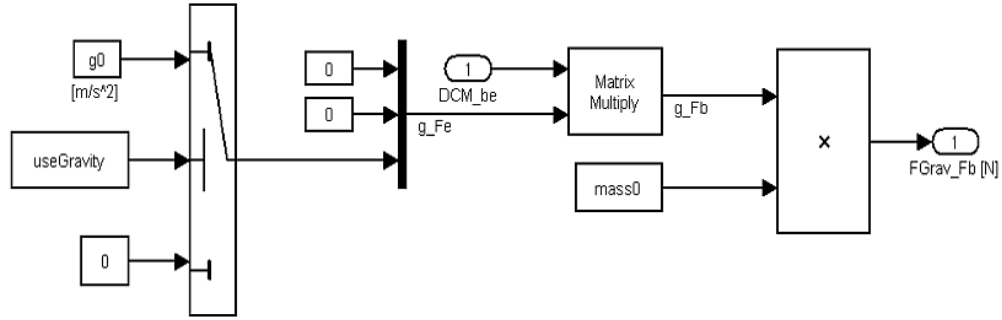


Figure 4. Gravitational force calculation.

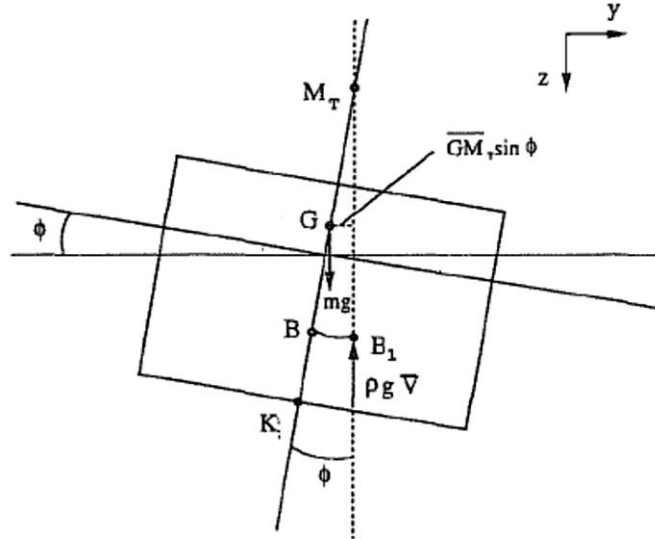


Figure 5. Hydrostatic forces and moments [15].

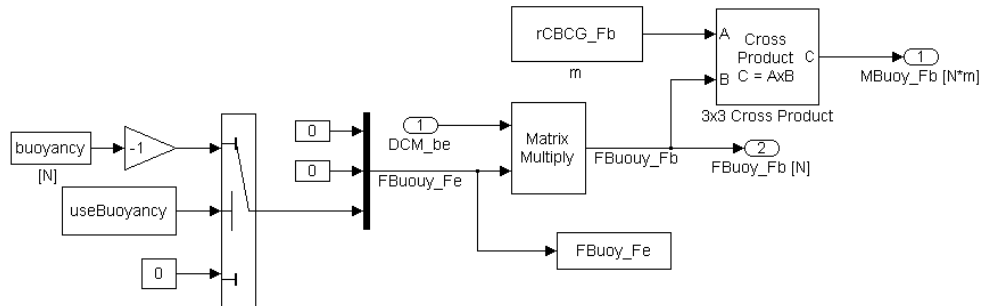


Figure 6. Hydrostatic force and moment calculation.

Similarly, after applying the hydrostatic moment coefficients to Equation (26), the moment equations become:

$$\begin{aligned} M_x &= I_x \dot{p} + (z_g W - z_b B) \cos \theta \sin \phi \\ M_y &= I_y \dot{q} - (I_z - I_x) r p + (z_g W - z_b B) \sin \theta \\ M_z &= I_z \dot{r} - (I_x - I_y) p q \end{aligned} \quad (31)$$

In Figure 6, hydrostatic forces and moments are implemented in Simulink.

2.1.8. Hydrodynamic Forces and Moments

In this subsection, the main hydrodynamic effects acting on an AUV moving in a fluid (seawater) will be briefly discussed. This is the key part of AUV modeling. Standard EOM contain only stability derivatives for the specific AUV configuration of interest.

Hence, trajectory simulation or prediction using traditional methods requires a priori knowledge of the hydrodynamic characteristics of the vehicle in the flow regimes, which may occur during the maneuver [15].

All hydrodynamic parameters are defined uniquely for a given AUV shape by formulae based on the results of theory and experiment.

There are primarily three methods of determining the hydrodynamic coefficients in the design process of underwater vehicles: (1) towing tank tests, (2) numerical computations, and (3) field tests. Among these methods, the most reliable results are obtained from the field tests, where the whole designed model is tested in a real sea environment [1]. Towing tank tests are performed with a scaled model and the hydrodynamic forces and moments can accurately be determined. However, experimental testing of designs is a time consuming and costly process (construction of the models, instrumentations, test infrastructure, etc.).

Numerical Computations are mainly based on semi-empirical or CFD methods. Semi-empirical or potential theory-based methods are generally utilized in the preliminary design process, where it is important to determine the hydrodynamic characteristics in a short period of time. CFD methods give accurate results and are used in the detailed design process [12].

The theory of fluid dynamics is rather complex and it is difficult to develop a reliable and robust model for most of the hydrodynamic effects. A rigorous analysis for incompressible fluids would need to resort to the *Navier-Stokes* equations, which are the basic governing equations for a viscous, heat-conducting fluid.

These equations describe how the velocity, pressure, temperature and density of a moving fluid are related. The hydrodynamic prediction method is coupled with a 6-DOF EOM solver to predict vehicle trajectories. The predicted motion characteristics of the AUV are also sensitive to details of the predicted hydrodynamic characteristics of the vehicle.

Small perturbations in the flow field, which cause small; variations in the vehicle forces and moments, accumulate over the length of the hull, and can produce large perturbation; in the calculated trajectory.

In this chapter in order to determine hydrodynamic forces and moments, we used *SUBFLO_2*, which is an engineering physically based and commercially available software tool. *SUBFLO_2* has four major components which are hull separation vortex method, fin horseshoe vortex method, propeller/propulsion models, and 6-DOF equations of motion

solver. The hydrodynamic prediction method is coupled with a 6-DOF EOM solver to predict vehicle trajectories. The predicted motion characteristics of the AUV are also sensitive to details of the predicted hydrodynamic characteristics of the vehicle. Small perturbations in the flow field, which cause small variations in the vehicle forces and moments, accumulate over the length of the hull, and these can produce large perturbation; in the calculated trajectory [23].

In order to determine hydrodynamic coefficients utilizing with *SUBFLO_2*, first we find three static hydrodynamic force coefficients of CF_x , CF_y , and CF_z , and then three static hydrodynamic moment coefficients of $CMSFx$, $CMSFy$, and $CMSFz$, for the predefined AUV geometry.

These coefficients are calculated due to the parameters of δ_s , δ_r , α , β , $V_{b/w}$, which are elevator deflection, rudder deflection, angle of attack, side slip angle, and velocity vector with respect to water, respectively. In Figure 7, the Simulink model of hydrodynamic force and moment calculation is shown.

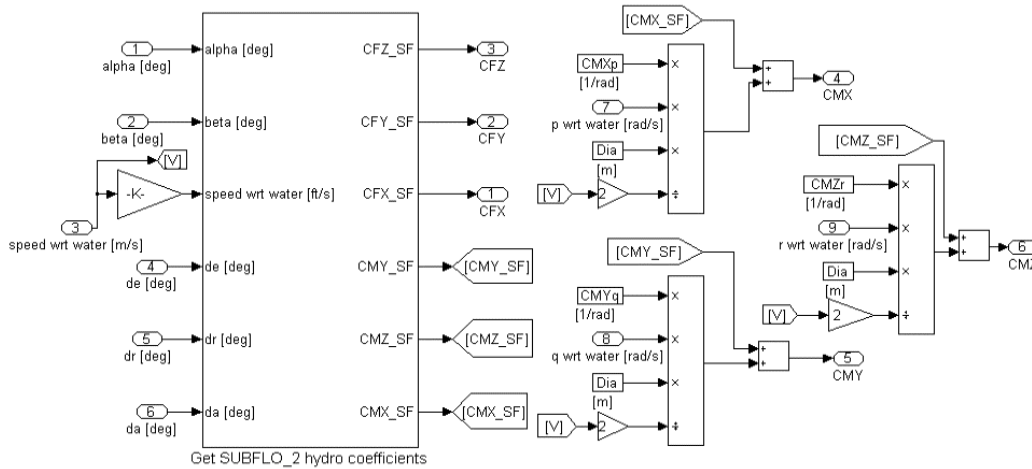


Figure 7. Hydrodynamic force and moment calculation.

2.1.9. Added Mass and Inertia

When a rigid body is moving in a fluid, the additional inertia of the fluid surrounding the body, which is accelerated by the movement of the body, has to be considered. This effect can be neglected in industrial robotics since the density of the air is much lighter than the density of a moving mechanical system.

However, in underwater applications the density of the water, $\rho \approx 1000 \text{ kg/m}^3$, is comparable with the density of the vehicles. In particular, at zero degrees, the density of the fresh water is 1002.68 kg/m^3 ; for sea water with 3.5% of salinity it is $\rho = 1028.48 \text{ kg/m}^3$.

Since the fluid surrounding the body is accelerated with the body itself, a force is then necessary to achieve this acceleration (the fluid exerts a reaction force which is equal in magnitude and opposite in direction).

This reaction force is the added mass contribution [16].

The added mass is not a quantity of fluid to add to the system such that it has an increased mass. Different properties hold with respect to the six by six inertia matrix of a rigid body due to the fact that the added mass is a function of the body's surface geometry.

As an example, the inertia matrix is not necessarily positive definite. The added mass has also an added *Coriolis* and *Centripetal* contribution.

In this study, “*Added Mass and Inertia*” effects of water is not taken into consideration. Added (virtual) mass should be understood as pressure-induced forces and moments due to a forced harmonic motion of the body, which are proportional to the acceleration of the body [12].

Consequently, the added mass forces and acceleration will be 180° out of phase to the forced harmonic motion. However, this is not true when AUV is close to surface.

2.1.10. Propeller Effect

The propeller produces the main thrust. Consequently, the reaction of the body to the load torque of the propeller produces a moment with respect to its rotational axis. The vehicle is a nonlinear system: all equations of motion of the system include coupled terms [1]. The main terms of this type are in the longitudinal (X) force and roll moment (K) equations because the thrust forces and moments act in the direction of the x-axis.

A propeller with a rudder can produce a thrust vector within a range of directions and magnitudes in the horizontal plane for low speed maneuvering and dynamic positioning. In our physically based model, we used a simple fixed-trust model due to the AUV velocity.

2.2. Mathematical Model of AUV

Detailed explanations on the theory behind the main concepts that directly influenced the design of the dynamic mathematical model of AUV are given in the previous sub-section. Now we arrive at the combined overall non-linear EOM for the AUV in 6-DOF as follows [18]:

Surge or translational motion along the x-axis:

$$\begin{aligned} X = & X_{\dot{u}}\dot{u} + m[-\dot{u} - z_G\dot{q} + y_G\dot{r}] + X_{uu}u|u| + (X_{wq} - m)wq + (X_{qq} + mx_g)q^2 + \\ & + (X_{vr} + m)vr + (X_{rr} + mx_g)r^2 - mz_g pr - (W - B)\sin\theta + X_{prop} \end{aligned} \quad (32)$$

Sway or translational motion along the y-axis:

$$\begin{aligned} Y = & Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + m[-\dot{v} + z_g\dot{p} - x_g\dot{r}] + Y_{uv}uv + (Y_{wp} + m)wp + (Y_{ur} - m)ur - \\ & - (mz_g)qr + (Y_{pq} - mx_g)pq + Y_{vv}v|v| + Y_{rr}r|r| + (W - B)\cos\theta\sin\phi + Y_{uu\delta_r}u^2\delta_r \end{aligned} \quad (33)$$

Heave or translational motion along the z-axis:

$$\begin{aligned} Z = & Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + m[-\dot{w} + x_g\dot{q} - y_g\dot{p}] + (Z_{uq} + m)uq + (Z_{vp} - m)vp + (mz_g)p^2 + \\ & + Z_{uw}uw + (mz_g)q^2 + (Z_{rp} - mx_g)rp + Z_{ww}w|w| + Z_{qq}q|q| + (W - B)\cos\theta\cos\phi + Z_{uu\delta_e}u^2\delta_e \end{aligned} \quad (34)$$

Roll or rotation about the x-axis:

$$\begin{aligned}
K = & mz_g \dot{u} - my_g \dot{r} - (I_{xx} - K_{\dot{p}}) \dot{p} + (my_g) \dot{w} - (I_{zz} - I_{yy}) qr - (mz_g) wp + \\
& + (mz_g) ur - (z_g W - z_b B) \cos \theta \sin \psi - (y_g W - y_b B) \cos \theta \cos \psi + K_{pp} p |p| + K_{prop}
\end{aligned} \quad (35)$$

Pitch or rotation about the y-axis:

$$\begin{aligned}
M = & -mz_g \dot{u} + (mx_g + M_{\dot{w}}) \dot{w} - (I_{yy} - M_{\dot{q}}) \dot{q} + (M_{rp} + I_{zz} - I_{xx}) rp + (mz_g) vr - \\
& - (mz_g) wq + (Muq - mx_g) uq + M_{uw} uw + (M_{vp} + mx_g) vp + M_{qq} q |q| + M_{ww} w |w| - \\
& - (z_g W - z_b B) \sin \theta - (x_g W - x_b B) \cos \theta \cos \psi + M_{uu\delta_e} u^2 \delta_e
\end{aligned} \quad (36)$$

Yaw or rotation about the z-axis:

$$\begin{aligned}
N = & my_g \dot{u} + (N_{\dot{v}} - mx_g) \dot{v} + (N_{\dot{r}} - I_{zz}) \dot{r} + (Npq + I_{xx} - I_{yy}) pq + (N_{wp} - mx_g) wp + \\
& + (N_{ur} - mx_g) ur + N_{uv} uv + N_{rr} r |r| + N_{vv} v |v| + (x_g W - x_b B) \cos \theta \sin \psi + (y_g W - y_b B) \sin \theta + N_{uu\delta_r} u^2 \delta_r
\end{aligned} \quad (37)$$

where X , Y , Z , K , M , and N represent the resultant forces and moments with respect to the body-fixed coordinates. The parameters of the mathematical model in Equations (32)-(37) with their definitions can be found in Tables 3-5 at the end of this chapter.

2.3. Numerical Integration of the AUV EOM

The nonlinear differential equations defining the AUV accelerations and the kinematic equations give us the vehicle accelerations in two different reference frames. Given the complex and non-linear nature of these equations, we will use numerical integration to solve for the vehicle speed, position, and attitude in time frame [19]. Consider that at each time step, we can express nonlinear differential equation as follows:

$$\dot{x}_n = f(x_n, u_n) \quad (38)$$

where x is the AUV state vector:

$$x = [u \ v \ w \ p \ q \ r \ x \ y \ z \ \phi \ \theta \ \psi]^T \quad (39)$$

and u_n is the input vector:

$$x = [u \ v \ w \ p \ q \ r \ x \ y \ z \ \phi \ \theta \ \psi]^T \quad (40)$$

There are two common numerical iteration methods to solve the non-linear differential equation: one is Euler's method and the other is Runge-Kutta method.

2.3.1. Euler's Method First Order

We will first consider Euler's method, a simple numerical approximation that consists of applying the iterative formula:

$$x_{n+1} = x_n + f(x_n, u_n) \cdot \Delta t \quad (41)$$

where Δt is the modeling time step.

Although the least computationally intensive method, Euler's method is unacceptable as it can lead to divergent solutions for large time steps [19].

2.3.2. Runge-Kutta Method

In this subsection, we will introduce one of the most powerful predictor-corrector algorithms- one which is so accurate, that most computer packages designed to find numerical solutions for differential equations will use it by default - the Runge-Kutta fourth order method. The Runge-Kutta method further improves the accuracy of the approximation by averaging the slope at four points [19]. We used this method to solve the nonlinear dynamic model of the AUV. In this method, numerical approximations that consist of applying the iterative formula:

$$\begin{aligned} k_1 &= f(x_n, u_n) \\ k_2 &= f\left(x + \frac{\Delta t}{2} k_1, u_{n+\frac{1}{2}}\right) \\ k_3 &= f\left(x + \frac{\Delta t}{2} k_2, u_{n+\frac{1}{2}}\right) \\ k_4 &= f(x + \Delta t k_3, u_{n+1}) \end{aligned} \quad (42)$$

where the interpolated input vector is:

$$u_{n+\frac{1}{2}} = \frac{1}{2}(u_n + u_{n+1}) \quad (43)$$

The combination of these two Equations (42)-(43) yields to:

$$x_{n+1} = x_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (44)$$

which is simply the x -value of the current point plus a weighted average of four different x -jump estimates for the interval, with the estimates based on the slope at the midpoint being weighted twice as heavily as those using the slope at the end-points [10].

For estimations of rigid-body aerodynamic or hydrodynamic models, the second order Runge-Kutta algorithm is usually adequate and hence recommended during the initial iterations of the iterative estimation algorithms, switching over the forth-order Runge-Kutta only during the final iteration [20].

2.4. Dynamic Simulink Model and Control of AUV

The data achieved from the modeling and identification process of the AUV was implemented into a vehicle which was used in a MATLAB Simulink environment.

This makes it possible to simulate the behaviour of the AUV or programs without the need of a real sea environment.

The detailed AUV Simulink models are depicted in Figure 8. In this model, hydrodynamic, hydrostatic, gravitational, and kinematic EOM of AUV and sliding mode control are implemented in the MATLAB version 7.5 Simulink environment.

The sliding mode control algorithm, which is a robust control, is applied to yaw channel of the dynamic model. The control input of course is

$$u = \psi - \Delta\psi - \lambda(r - \Delta\dot{\psi}) \quad (45)$$

The sliding mode control law for course control of AUV is

$$\frac{-5\pi}{18}|u|sign(u) \quad (46)$$

The control strategy used for physically based model of AUV is depicted in Figure 9, which is the Simulink model of sliding mode control of yaw (ψ).

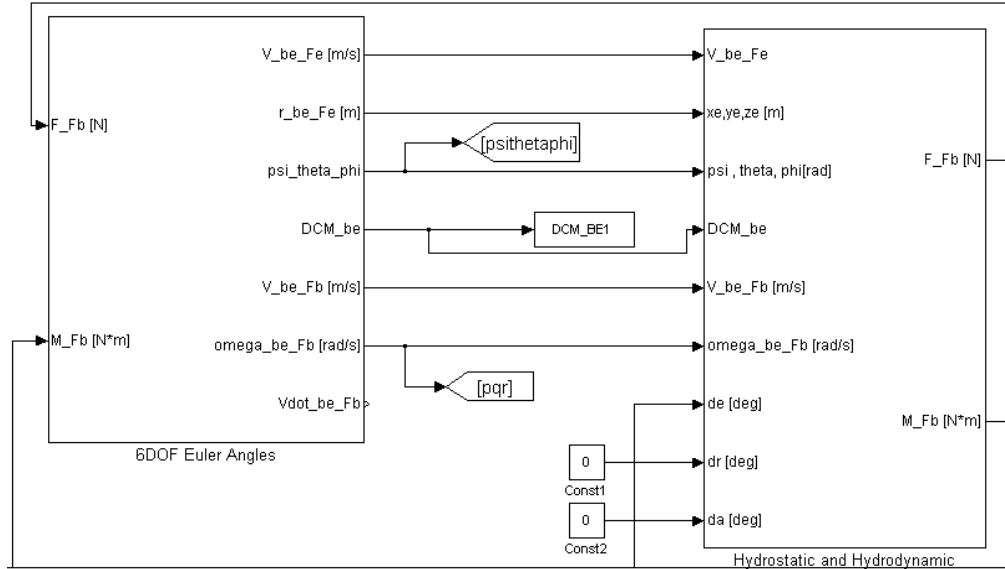


Figure 8. Simulink model of AUV.

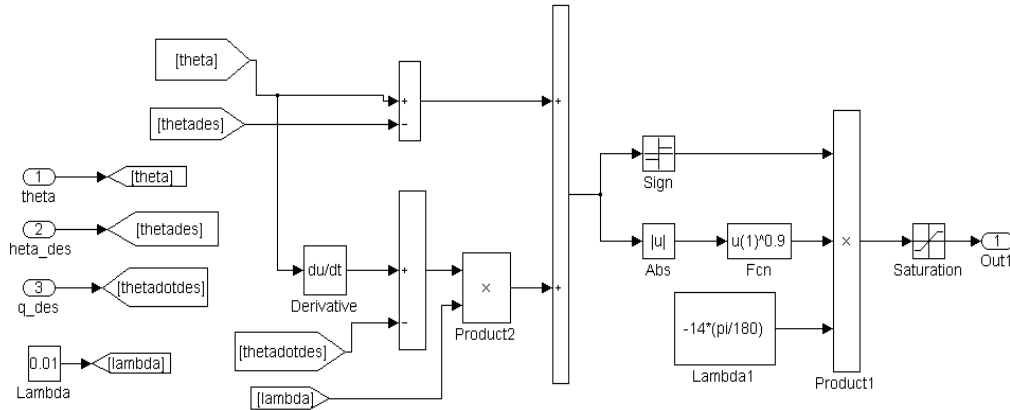


Figure 9. Simulink Model of Sliding Mode Control.

2.5. Simulation Results

The simulation of the dynamic model is developed by using the MATLAB version 7.5 Simulink environment. Numerical simulations are made to show the dynamic model of the AUV.

In Figure 10, the simulation results of depth change in z axis, dynamic position maneuver in x and y axis, and yaw angle change are shown. In Figure 16, the simulation result of deflection angle change of rudder under the sliding mode control and total velocity during 50 sec. For this simulation, the AUV has a maneuver speed of 8 m/s and rudder angle applies to 10 deg. From the start, elevator angle applies to 1 deg, and the simulation time is 50 sec.

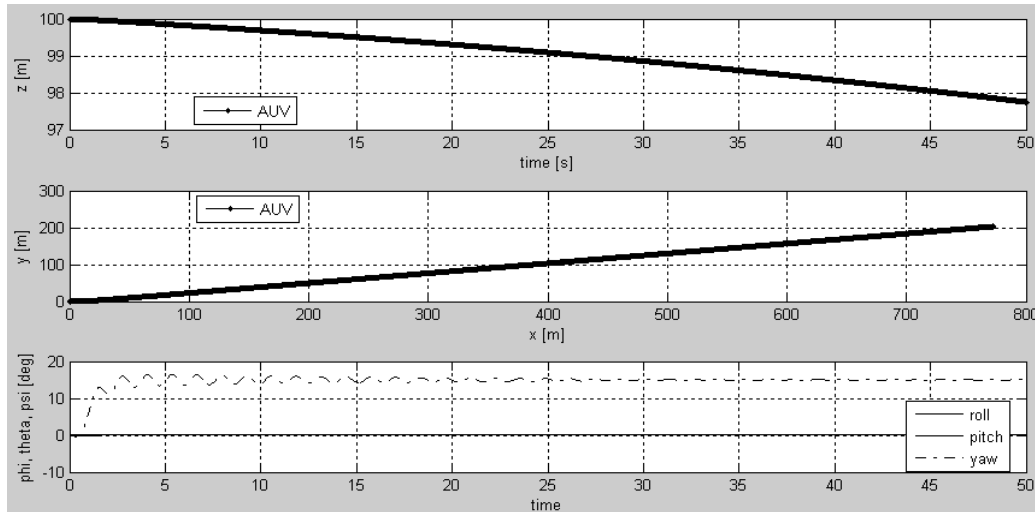


Figure 10. Simulation Results of X, Y, Z, Roll, Pitch and Yaw Angles (50 sec).

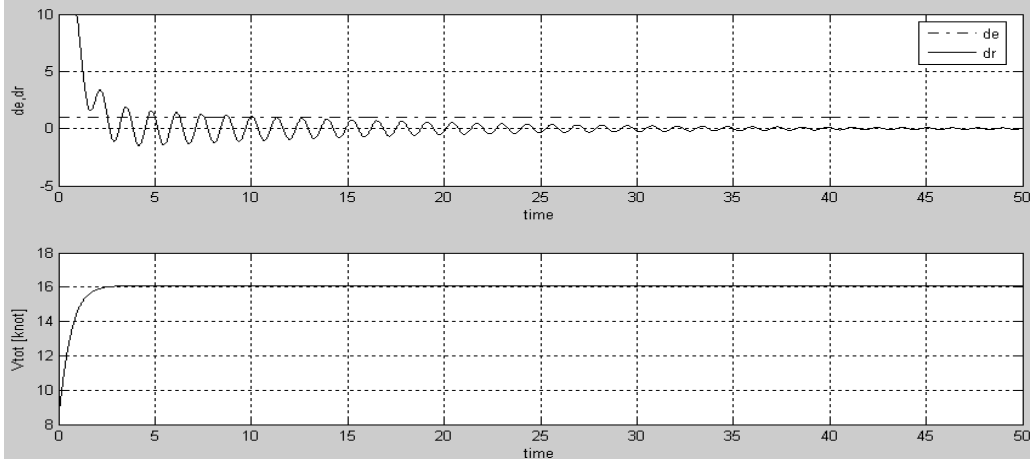


Figure 11. Simulation Results of V_{tot} , δr and δe (50 sec).

3. PARAMETER IDENTIFICATION WITH LEAST SQUARES ESTIMATION

This section focuses on the parameter identification of hydrodynamic coefficients of AUVs based on the *Least Square Estimation* (LSE) algorithm for a nonlinear mathematical modeling of AUVs. Hydrodynamic coefficients strongly affect the dynamic performance of an AUV. Therefore, it is important to have the true values of these coefficients in order to accurately simulate the AUV's dynamic performance. The estimated coefficients can be used as inputs not only for a mathematical model to analyze the maneuvering performance but also for a controller model to design AUVs under development.

However, parameter identification of AUV dynamics is complicated because of its nonlinear identification models and the combination of noisy sensor measurements.

3.1. AUV System Identification

System identification is the determination on the basis of an observation of input and output, of a system within a specified class of systems to which the system under test is equivalent [20].

When the system identification is applied to AUVs, the equations governing the AUV dynamic motion are postulated and an experiment is designed to obtain measurements of input and output variables [21].

The hydrodynamic forces and moments acting on AUV cannot be measured directly.

However, hydrodynamic modeling followed by parameter estimation allows determination of specific hydrodynamic characteristics (such as lift, drag, and side force coefficients, and rolling, pitching, and yawing moment coefficients in terms of stability and control derivatives) from the related measurements such as accelerations, angular rates, etc. [20].

The selected EOM for system identification are sway and yaw

$$\begin{aligned}
Y = & Y_v \dot{v} + Y_r \dot{r} + m \left[-\dot{v} + z_g \dot{p} - x_g \dot{r} \right] + Y_{uv} uv + (Y_{wp} + m) wp + (Y_{ur} - m) ur - \\
& - (m z_g) qr + (Y_{pq} - m x_g) pq + Y_{vv} v |v| + Y_{rr} r |r| + (W - B) \cos \theta \sin \phi + Y_{uu \delta_r} u^2 \delta_r
\end{aligned} \tag{47}$$

$$\begin{aligned}
N = & m y_g \dot{u} + (N_v - m x_g) \dot{v} + (N_r - I_{zz}) \dot{r} + (N p q + I_{xx} - I_{yy}) p q + (N_{wp} - m x_g) wp + (N_{ur} - m x_g) ur + \\
& + N_{uv} uv + N_{rr} r |r| + N_{vv} v |v| + (x_g W - x_b B) \cos \theta \sin \psi + (y_g W - y_b B) \sin \theta + N_{uu \delta_r} u^2 \delta_r
\end{aligned} \tag{48}$$

The dynamic Equations (47) and (48) are augmented with output equations that specify the connection of AUV states and controls to measured outputs, along with measurement equations describing the measurement process [21]. In Figure 12, *Model Postulation* is based on a priori knowledge about the AUV kinematics and hydrodynamics.

The *Experiment Design* includes selection of an instrumentation system, and specification of an AUV configuration and maneuvers for system identification.

Data Compatibility Analysis, in practice, measured AUV response data that can contain systematic errors, even after careful instrumentation and experimental procedures.

To verify data accuracy, data compatibility analysis can be applied to measured AUV responses. *Model Structure Determination* in AUV system identification means selecting a specific form for modeling from a class of models, based on measured data. *Parameter and State Estimation* constitutes a principal part of the AUV system identification procedure. Parameter estimation for linear dynamic systems based on maximum likelihood and the least squares principles can also be formulated in the frequency domain.

Collinearity diagnostics is in almost all practical applications of linear regression and the model terms are correlated to some extent. Diagnostic information can aid in deciding what corrective actions are necessary.

Model Validation is the last step in the identification process [22]. The identified model must demonstrate that its parameters have physically reasonable values and acceptable accuracy, and that model has a good prediction capability on comparable maneuvers [21].

3.2. Parameter Estimation for AUV

The parameters can be estimated from calculating the first principles of dynamics of AUVs, from statistical LSE of experimental data, or a combination of the two. The identification of the parameters of the item being tracked can be approached in many different ways. The parameters can be determined once, from a model of the expected input signal.

It also could be done adaptively utilizing a recursive method to identify the parameters of the incoming signal.

The modern era of system identification is marked by the implementation of the maximum-likelihood method [20]. Therefore, we used the *Least Square model* for the uncertainties in the parameters and measurements in this chapter. Based on this model, is a vector of unknown constant parameters and v is a random vector of measurement noise. There are two different models that are Bayesian and Fisher [21]. The LSE algorithm always gives the best linear fit when the noise is white and Gaussian.

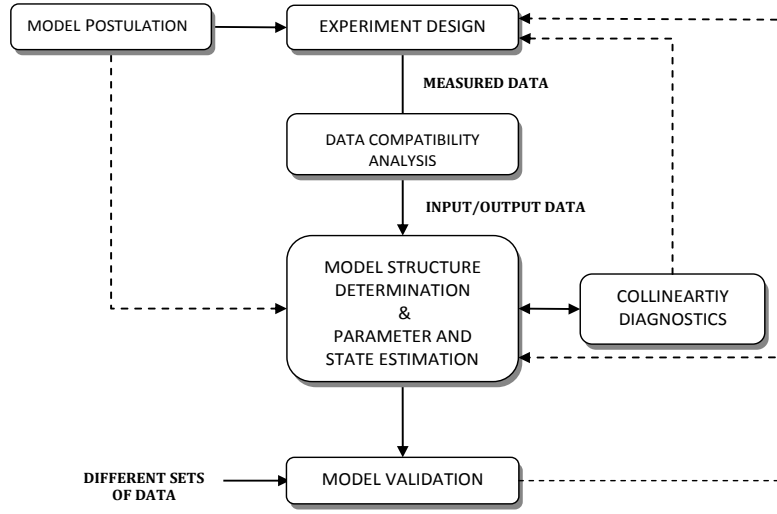


Figure 12. AUV System Identification Method [21].

Moreover, it is easier to tune the identification process in the latter part with a priori knowledge of the system [21]. A recursive LSE was chosen in this study for the parameter estimation since it is easy to apply the system dynamic and allows for the consideration of modeling and measurement errors.

The parameter estimation process consists of finding values of unknown model parameter θ in an assumed model structure, based on noisy measurements z . An estimator is a function of the random variable z that produces an estimate $\hat{\theta}$ of the unknown parameters θ . Since the estimator computes $\hat{\theta}$ based on noisy measurements z , $\hat{\theta}$ is a random variable [21]. In order to estimate the hydrodynamic coefficients of AUVs, the LSE is designed using the observer model. The estimated parameters yields to

$$\theta = [Y_{vv} Y_{rr} Y_{uv} Y_{ur} Y_{wp} N_{vv} N_{rr} N_{uv} N_{ur} N_{wp}] \quad (49)$$

The output y is given by:

$$y = X\theta \quad (50)$$

where the matrix X is assumed to be known. Then the measurement equation can be expressed as:

$$z = X\theta + v \quad (51)$$

where $z = [z(1) z(2) \dots z(N)]^T : N \times 1$ is the length measurement vector, $\theta = [\theta_0 \theta_1 \dots \theta_n]^T : n_p \times 1$ is the vector of unknown parameters ($n_p = n+1$), $\theta = [\theta_0 \theta_1 \dots \theta_n]^T : n_p \times 1$ is the matrix of vectors of ones and regressors, and $v = [v(1) v(2) \dots v(N)]^T : N+1$ is the vector of measurement errors.

In general, there are n_0 measured outputs, and a vector of measurements is taken at each sample i , where $i = 1, 2, 3, \dots$, N is the number of sampled data points. A single measured output is assumed, $son_0=1$, and z is a vector composed of N scalar measurements [21].

Neglecting interactions from other parts of the system, the sway and yaw measurement subsystems can be modeled as

$$Z_{Ymeas} = Y - mwp + (mz_g)qr + (mx_g)pq - Y_{uu\delta_r} u^2 \delta_r \quad (52)$$

$$Z_{Nmeas} = N - (N_{pq} + I_{xx} - I_{yy})pq + (y_g W) \sin \theta + N_{uu\delta_r} u^2 \delta_r \quad (53)$$

For the *least squares model*, there are no probability statements regarding θ , but v is assumed to be zero mean and uncorrelated, with a constant variance:

$$E(v) = 0 \quad \text{and} \quad E(vv^T) = \sigma^2 I \quad (54)$$

The unknown parameter vector θ can be estimated via LSE as follows

$$\hat{\theta} = (X^T X)^{-1} X^T z \quad (55)$$

The $n_p \times n_p$ matrix $X^T X$ matrix is always symmetric. If the regressor vectors that make up the columns of X are linearly independent, then $X^T X$ is positive definite and the eigenvalues of $X^T X$ are positive real numbers, and the associated eigenvectors are mutually orthogonal so the $X^T X$ exists [21].

Finally, the covariance matrix of the parameter estimate $\hat{\theta}$ also known as the covariance matrix of the estimation error $\hat{\theta} - \theta$ is simplified to

$$\text{cov}(\theta_{est}) \equiv E[(\theta_{est} - \theta)(\theta_{est} - \theta)^T] = \sigma^2 (X^T X)^{-1} \quad (56)$$

3.3. Simulation Results of Parameter Identification

Numerical simulations are made to show the parameter identifications of the AUV. The hydrodynamic coefficients associated with horizontal and vertical motions are estimated by simulating the combined yaw and sway motion of the AUV.

The numerical values of the AUV physical parameters, hydrodynamic force coefficients and hydrodynamic moment coefficients are given in Tables 3-5 respectively.

We will describe how state-of-the art algorithms are used to find a dynamic model that enables accurate simulation of steering dynamics.

In summary, the system identification task to estimate the parameters (56). The parameter identification results are given in Table 2.

Numerical results demonstrate that the LSE estimates the hydrodynamic coefficients very well.

Table 2. Parameter Identification Results

Parameters	Real Value	Est. Value with Int. Nav.	Percent %	Difference
Y_{vv}	-196,26	-192,650	1,839	3,610
Y_{rr}	8,30	8,580	3,373	0,280
Y_{uv}	38,39	38,277	0,295	0,113
Y_{ur}	13,41	13,320	0,674	0,90
Y_{wp}	35,50	36,105	1,704	0,605
N_{vv}	-7,95	-7,733	2,736	0,218
N_{rr}	-24,13	-24,051	0,328	0,079
N_{uv}	-21,89	-21,843	0,216	0,047
N_{ur}	1,93	1,922	0,435	0,008
N_{wp}	-19,30	-19,367	0,347	0,067

The simulation results of the real model and identified model are shown in Figures 13-15 respectively. Figures show that the AUV had a surge velocity changed between 5 and 0 m/s. The depth changed between -15 m and -13.5 m. The elevator was operated at the maximum deflection of 15 deg for most of the time and similarly the rudder deflection operated at the maximum deflection of -11 deg. The iteration number N is chosen to be 5,000 and the sampling time is chosen to be 0.001 sec. As seen from Figures 13, 14 and 15, outputs of real and identified models are very close. On the one hand, this is obviously seen in Figure 16. As a result, Simulation results show that the proposed method works very well. For model verification, the existing in literature methods [10, 20, 22] can be used.

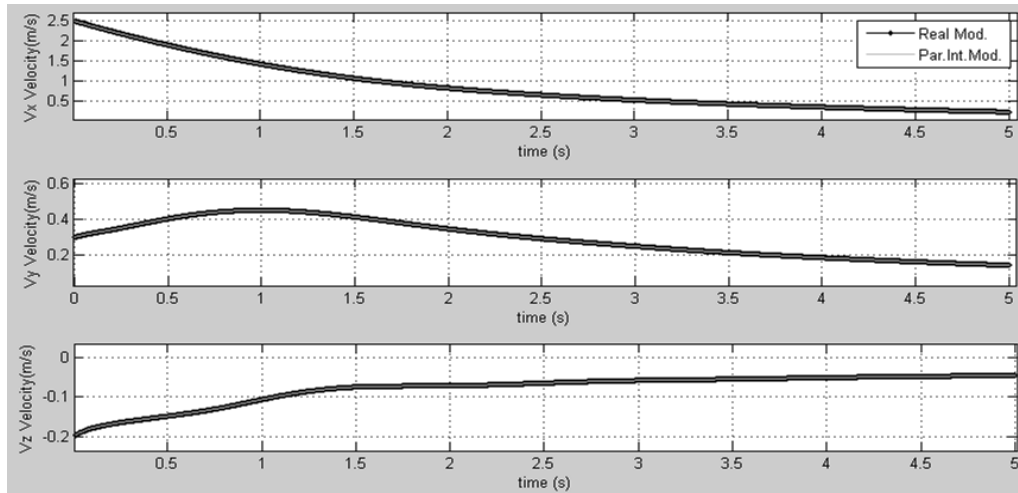


Figure 13. Velocity simulation results of real and identified model.

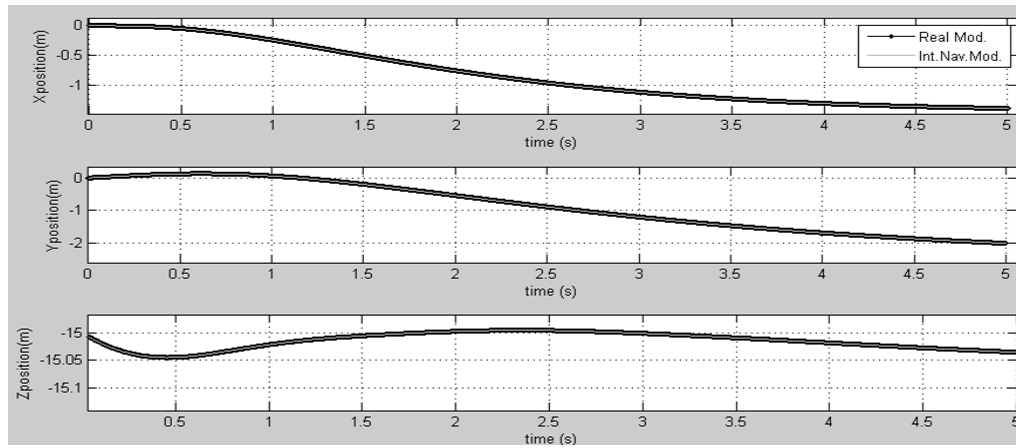


Figure 14. Position simulation results of real and identified model.

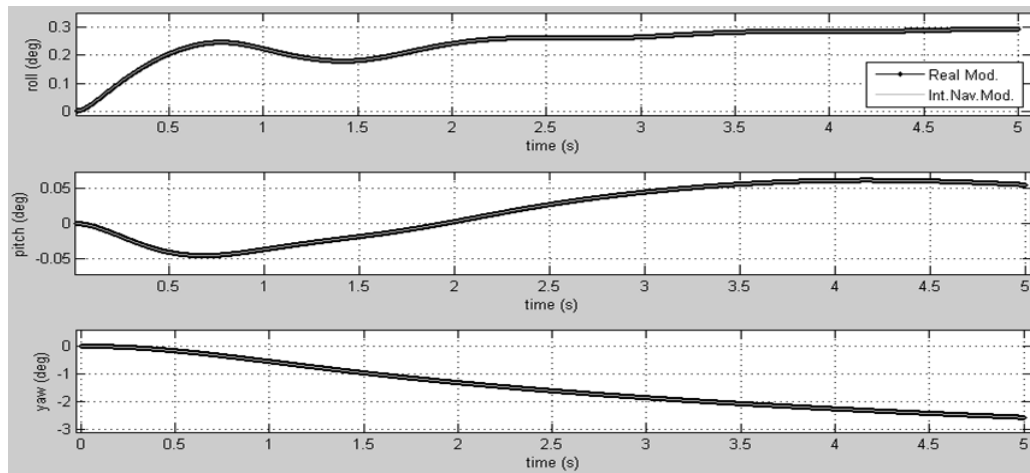


Figure 15. Attitude simulation results of real and identified model.

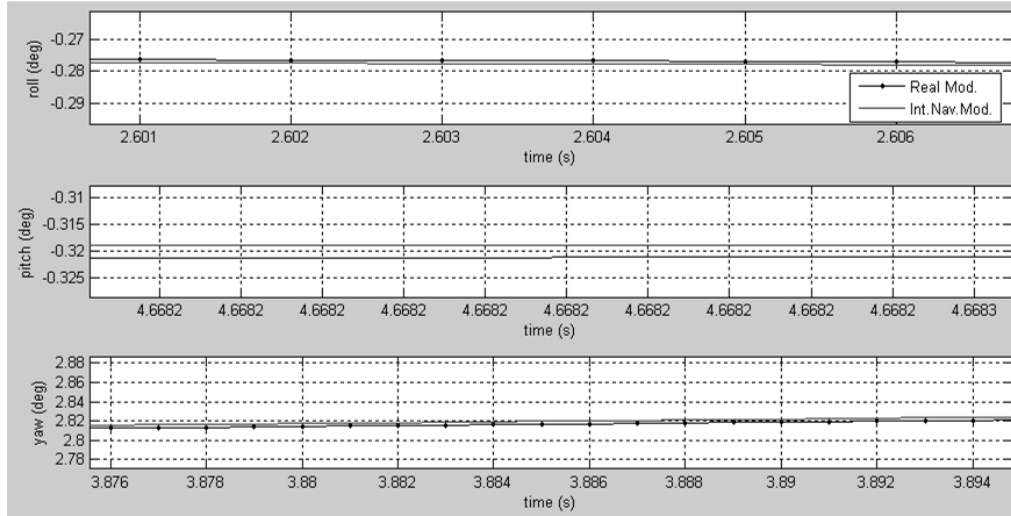


Figure 16. Attitude simulation results of real and identified model (Zoom in Version).

Table 3. AUV Physical Parameters [10]

Parameter	Value	SCI unit	Definition
ρ	1010	g/m^3	Fluid density
g	9.81	m/s^2	Gravitational acceleration
d	0.191	m	Hull cylindrical radius
l	1.33	m	Overall hull length
S_w	0.7981	m^2	Submerged Area (πdl)
A_p	0.2540	m^2	Body base area (dl)
A_f	0.0287	m^2	Body cross area ($\pi d^2/4$)
c_{ds}	0.004	-	Surface drag coefficient
c_{dF}	0.3	-	Drag coefficient (A_f)
c_{da}	0.0166	-	Total drag coefficient (for S_w)

Table 3. (Continued)

Parameter	Value	SCI unit	Definition
l_{cp}	0.2645	M	Distance to center of pressure
t_{fin}	1010	g/m^3	Fluid density
h_{fin}	0.0960	m	Rudder hight
A_r, A_s	0.0071	m^2	Vertical/horizontal rudder surface area
c_{df}	1.558	-	Rudder drag coefficient
W	300	N	AUV weight
B	306	N	Bouyant force
x_b	0	m	Bouyancy center about the x-axis
y_b	0	m	Bouyancy center about the y-axis
z_b	0	m	Bouyancy center about the z-axis
x_g	0	m	Gravity center about the x-axis
y_g	-0.008	m	Gravity center about the y-axis
z_g	0.0196	m	Gravity center about the z-axis

I_{xx}	0.177	kg m ²	Moment of inertia about the x-axis
I_{yy}	3.45	kg m ²	Moment of inertia about the y-axis
I_{zz}	3.45	kg m ²	Moment of inertia about the z-axis
R	5.87E-2	m	Diameter of propeller
u_0	1.5	m/s	Design velocity

Table 4. AUV Hydrodynamic Force Coefficients [10]

Parameter	Value	SCI unit	Definition
$X_{u u }$	-6.68	kg/m	Axial drag
$X_{\dot{u}}$	-0.513	kg	Added mass
$Y_{v v }$	-196.26	kg/m	Cross flow drag
$Y_{r r }$	8.30	kg m/rad ²	Cross flow drag
Y_{uv}	-38.39	kg/m	Body and rudder bouyancy
$Y_{\dot{v}}$	-42.13	kg	Added mass
$Y_{\dot{r}}$	-5.16	kg m/rad	Added mass
Y_{ur}	13.41	kg/rad	Added mass + Rudder bouyancy
$Y_{uu\delta_r}$	16.99	kg/m/rad	Bouyant force of rudder
$Z_{w w }$	-196.26	kg/m	Cross flow drag
$Z_{q q }$	-8.30	kg m/rad ²	Cross flow drag
Z_{uw}	-38.39	kg/m	Body and rudder bouyancy
$Z_{\dot{w}}$	-42.13	kg	Added mass
$Z_{\dot{q}}$	5.16	kg m/rad	Added mass
Z_{uq}	-13.41	kg/rad	Added mass + Rudder bouyancy
$Z_{uu\delta_s}$	-16.99	kg/m/rad	Rudder bouyancy

Table 5. AUV Hydrodynamic Moment Coefficients [10]

Parameter	Value	SCI unit	Definition
$K_{p p }$	-5.03	kg m ² /rad ²	Roll drag
$K_{\dot{p}}$	-0.095	kg m ² /rad ²	Added mass moment
$M_{w w }$	7.95	kg	Cross flow drag
$M_{q q }$	-24.13	kg m ² /rad ²	Cross flow drag
M_{uw}	21.89	kg	Added mass + Rudder bouyancy + MunkMomenti
$M_{\dot{w}}$	5.16	kg m	Added Mass Inetia
$M_{\dot{q}}$	-7.57	kg m ² /rad	Added Mass Inetia
M_{uq}	-16.56	kg m/rad	Added mass + Rudder bouyancy
$M_{uu\delta_s}$	-13.92	kg/rad	Rudder Bouyancy Moment

$N_{v v }$	-7.95	kg	Cross flow drag
$N_{r r }$	-24.13	kg m ² /rad	Cross flow drag
N_{uv}	-21.89	kg	Added mass + Rudder bouyancy + MunkMomeni
$N_{\dot{v}}$	-5.16	kg m	Added Mass Inertia
$N_{\dot{r}}$	-7.57	kg m ² /rad	Added Mass Inertia
$N_{uu\delta_r}$	-13.92	kg rad	Rudder Bouyancy Moment

CONCLUSION

The research on underwater systems has gained immense attention during the last two decades because of applications taking place in many fields. Therefore, the significant number of UUVs has been developed for solving the wide range of scientific and applied tasks of ocean research and development in the world. The military, as well as civilian industry can see great potential uses of AUVs in the underwater environment.

This chapter is concerned with implementing dynamic modeling of an AUV. This chapter has also given an insight and theoretical background about AUV kinematics, hydrodynamics and hydrostatic. Because physically based mathematical modeling of an AUV is an interdisciplinary study of kinematics, hydrostatics, and hydrodynamics.

To achieve an accurate hydrodynamic model for any vehicle that is not a standard shape by purely analytical means is extremely difficult at best.

This chapter also discussed the use of parameter identification methods to predict the hydrodynamic derivatives of the AUV. LSE is used to tackle the problem of parameter identification of an AUV. This classical approach seems to have a better performance in cases where a specific parameter will be identified and the LSE results obtained are satisfactory. Therefore, numerical results demonstrate that the proposed method works very well, in both cases of using simulated data. It has been shown that the characteristic functions for LSE can be obtained in a stable way even in noisy environments.

Numerical simulations are made to show the parameter identifications of the AUV. The hydrodynamic coefficients associated with horizontal and vertical motions are estimated by simulating the combined yaw and sway motion of the AUV.

The simulation results indicate that for AUV hydrodynamic parameter identification purposes, the LSE algorithm remains a feasible tool, which consistently returns quality results and is the least costly in terms of computational demand.

Autonomous guidance, navigation, and control techniques are key research and development areas for success of the specific AUV missions.

Moreover, further work is needed for in precision navigation, sensor development and integration, and improving the reliability and robustness of long term and complex mission completion.

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